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Indian Institute of Engineering Science and Technology, Shibpur

Department of Computer Science & Technology...

Report on 8085 Assembly language Programming Continued.



REPORT ON

Experiment No. 3

Title 8085 Assembly language
Programming (continued)

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Semester 5th Date of Submission 10.08.2026

Branch Computer Science Roll No. 2024CSB046 Signature _____
and Technology

Examined by _____



Report on

1. Title

Experiment No. 3: 8085 Assembly Language Programming (Continued)

2. Date

August 3, 2026

3. Objective

To write simple programs in 8085 Assembly language and test them at the 8085 SDK (System Development Kit).

4. Work Done

Task 1: Executed and Validated Example 1 (Alternative 1)
Result: Contents of memory location 8800H - 05H
Contents at ~~memory~~ location in Register A - 05H

Task 2: Executed and Validated Example 1 (Alternative 2)
Result: Contents of memory location 8800H - 05H

Task 3: Executed and Validated Example 2
Result: Contents of 8801H memory location - 00H

Task 4: Executed and Validated Example 3
Result: Contents of Register D - 0AH(10)

Task 5: Program 1 ($z = y + (2 * x > y) ? 10 : 5;$)
Action: Wrote and tested the assembly code. let $x = 5$
(05H at 8800H) and $y = 15$ (0FH at 8801H)



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Label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
Start:		8800	05	$x \leftarrow 05$
	Data Area {	8801	0F	$y \leftarrow 15$
		8802	--	z uninitialized
		8803	3A	$A \leftarrow x$
	LDA 8800	8804	00	
		8805	88	
	ADD A	8806	87	$A \leftarrow A + A (2 * x)$
	MOV B, A	8807	47	$B \leftarrow 2 * x$
	LDA 8801	8808	3A	$A \leftarrow y$
		8809	01	
		880A	88	
	CMP B	880B	B8	Compare y with $2 * x$
	JC Greater	880C	DA	if $y < 2 * x$, jump
		880D	14	
		880E	88	
	MVI A, 05	880F	3E	$A \leftarrow 05$
		8810	05	
	JMP CALC	8811	C3	Jump to 8816
		8812	16	
		8813	88	
	Greater MVI A, 0A	8814	3E	$A \leftarrow 10$
		8815	0A	
		8816	21	$HL \leftarrow 8801$
	CALC LXI H, 8801	8817	01	
		8818	88	
	ADD M	8819	86	$A \leftarrow y + A$



Report on

label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
	STA 8802	881A	32	8802 ← A
		881B	02	
		881C	88	
	RST 3	881D	DF	Stop

- Result: 14H is stored at 8802H.

Task 6: Program 2 ($z = x \& y$)

- Action: Write code for bitwise ANDing - let $x = 9$ (09H at 8800H) and $y = 7$ (07H at 8801H)

label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
Start	Data Area {	8800	09	$x \leftarrow 09$
		8801	07	$y \leftarrow 07$
		8802	--	z uninitialized
	LDA 8800H	8803	3A	Load x in A
		8804	00	
		8805	88	
	MOV B, A	8806	47	$B \leftarrow A$
	LDA 8801H	8807	3A	$A \leftarrow 8801$
		8808	01	
		8809	88	
	ANA B	880A	A0	$A \leftarrow A \text{ AND } B$
	STA 8802H	880B	32	8802 ← A
		880C	02	
		880D	88	
	RST 3	880E	DF	Stop

Result: 01H is stored at 8802H



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Task 7: ~~Program~~ Program 3 ($2 * x < y$)

Action: Wrote and tested the assembly code. let $x=5$ (05H at 8800H) and $y=15$ (0FH at 8801H)

Label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
Start	Data Area	8800	05	$x \leftarrow 05$
		8801	0F	$y \leftarrow 15$
		8802	--	z uninitialized
				$A \leftarrow y$
	LDA 8801	8803	3A	
		8804	01	
		8805	88	
	MOV B, A	8806	47	$B \leftarrow y$
	LDA 8800	8807	3A	$A \leftarrow x$
		8808	00	
		8809	88	
	ADD A	880A	87	$A \leftarrow 2 * x$
	CMP B	880B	B8	Compares A and B
	JC Greater	880C	DA	if $2 * x < y$ jump
		880D	14	
		880E	88	
	MVI A, 05	880F	3E	$A \leftarrow 05$
		8810	05	
	JMP CALC	8811	C3	Jump to 8816
		8812	16	
		8813	88	
				$A \leftarrow 10$
	Greater MVI A, 0A	8814	3E	
		8815	0A	$HL \leftarrow 8801$
	CALC LXI H, 8801	8816	21	
		8817	01	
		8818	88	
				$A \leftarrow y + A$
	ADD M	8819	86	



Report on

Label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
	STA 8802	881A	32	8802 ← A
		881B	02	
		881C	88	
	RST 3	881D	0F	Stop

- Result: 19H is stored at 8802H

Task 8: ~~Assign~~ Program 4 ($2^*x == y$)

- Action: Modified Assignment 1. SC gets replaced with SZ at 880CH, opcode = C2

- Result: 14H is stored at 8802H

Task 9: Program 5 ($2^*x != y$)

- Action: Modified Program 4. SZ gets replaced with JNZ at 880CH, opcode = C2.

- Result: 19H is present at 8802H.

Task 10: Program 6 ($2^*x >= y$)

- Action: Modified Program 1 conditional jump to evaluate greater than or equal to

- Result: 14H is stored at 8802H

Task 11: Program 7 ($2^*x <= y$)

- Action: Modified Program 1. SC gets replaced with SNC at 880CH, opcode = D2

- Result: 19H is stored at 8802H.

Task 12: Program 8 (Switch Case for Department)

- Action: Character mapped to 8800H, comparing ASCII values to store string in dept[10] at 8801H-880AH.



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Label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
	Data Area {	8800'	--	Memory for ch
		8801-880A	--	Memory for dept[10]
	LDA 8800	8000	2A	A ← ch
		8001	00	
		8002	88	
	CPI 41H	8003H	FE	Compare A with 'A'
		8004H	41	
	JZ CASEA	8005	CA	Jump if ch == 'A'
		8006	27	
		8007	80	
	CPI 43H	8008	F3	Compare A with 'C'
		8009	43	
	JZ CASEC	800AH	CA	Jump if ch == 'C'
		800B	2B	
		800C	80	
	LXI H, 8801	800D	21	HL ← dept
		800E	01	
		800F	88	
	MVI H, 49H	8010	36	Store 'I' in dept[0]
		8011	49	
	INX H	8012	23	HL++
	MVI M, 49	8013	36	Store 'I' in dept[1]
		8014	49	
	INX H	8015	23	HL++
	MVI M, 45	8016	36	Store 'E' in dept[2]
		8017	45	
	INX H	8018	23	HL++
	MVI M, 53	8019	36	Store 'S' in dept[3]
		801A	53	
		801B	23	HL++



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Label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
CASE-A	MVI H,54	801C	36	Store 'T' in dept[4]
		801D	54	
	INX H	801E	23	HL++
	MVI H,53	801F	36	Store 'S' in dept[5]
		8020	53	
	INX H	8021	23	HL++
	MVI H,00	8022	36	Store null in dept[6]
		8023	00	
	SHP END	8024	C3	Jump to End
		8025	409	
		8026	80	
	LXI H,8801	8027	21	HL ← dept[0]
		8028	01	
		8029	88	
	MVI H,41	802A	36	Store 'A' in dept[0]
		802B	41	
	INX H	802C	23	HL++
	MVI H,52	802D	36	Store 'R' in dept[1]
		802E	52	
	INX H	802F	23	HL++
				Store 'C' in dept[2]
	MVI H,43	8030	36	
		8031	43	
	INX H	8032	23	HL++
				Store 'H' in dept[3]
	MVI H,48	8033	36	
		8034	48	
	INX H	8035	23	HL++
				Store null in dept[4]
	MVI H,00	8036	36	
		8037	00	
	SHP END	8038	C3	Jump to End
		8039	409	
		803A	80	



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label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
CASE_C	LXI H, 8801	803B	21	HL ← dept[0]
		803C	01	
		803D	88	
	MVI H, 43	803E	36	Store 'c' in dept[0]
		803F	43	
	INX H	8040	23	HL++
	MVI M, 53	8041	36	Store 's' in dept[1]
		8042	53	
	INX H	8043	23	HL++
	MVI M, 54	8044	36	Store 't' in dept[2]
		8045	54	
	INX H	8046	23	HL++
	MVI M, 00	8047	36	Store null in dept[3]
		8048	00	
	RST 3	8049	DF	Stop

Result: By changing the value at 8800H to 41H ('A'), the ASCII values for "ARCH" (41H, 52H, 43H, 48H, 00H) were seen at memory locations starting from 8801H.

Task 13: Executed and Validated Example 4

Action: Executed subroutine passing parameters through 8800H and 8801H

Result: Checked memory location 8802H to verify the return value (the large number) was stored successfully.

Task 14: Program 9 (int isupper(char c))

Action: Write subroutine to check if a character is between 'A' (41H) and 'Z' (5AH).



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Label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
Parameters		8800H	—	Memory location for parameter C
isupper:	LDA 8800H	8A00	3A	$A \leftarrow C$
		8A01	00	
		8A02	88	
	CPI 41H	8A03	FE	Comp A with 'A'
		8A04	41	
	JC false	8A05	DA	$C < 'A' \rightarrow \text{Jump}$
		8A06	10	
		8A07	8A	
	CPI 5BH	8A08	FE	Comp A with 'Z'+1
		8A09	5B	
	JNC false	8A0A	D2	$C > 'Z'+1 \rightarrow \text{Jump}$
		8A0B	10	
		8A0C	8A	
	MVI B, 01H	8A0D	06	$B \leftarrow 1$
		8A0E	01	
	RET	8A0F	C9	Return to caller
false:	MVI B, 00H	8A10	06	$B \leftarrow 0$
		8A11	00	
	RET	8A12	C9	Return to caller
main:	MVI A, 43H	8900	3E	$A \leftarrow 'C' (\text{Test})$
		8901	43	
	STA 8800H	8902	32	Store at 8800H
		8903	00	
		8904	88	
	CALL isupper	8905	CD	Call subroutine
		8906	00	



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Label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
	MOV A,B	8907	8A	A ← B (return val) Store at 8802H
	STA 8802H	8908	78	
		8909	32	
		890A	02	Stop
		890B	88	
	RST 83	890C	DF	

-Result: The test character 'C' (43H) is uppercase.
After execution, checking memory location 8802H yields 01H.

Task 15: Program 10 (int islower (char c))

-Action: Write subroutine to check if a character is between 'a' (61H) and 'z' (7AH)

Label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
Parameters		8800H		Memory location for parameter C
islower:	LDA 8800H	8B00	3A	A ← C
		8B01	00	
		8B02	88	
	CPI 61H	8B03	FE	Comp A with 'a'
		8B04	61	
	JC false2	8B05	DA	C < 'a' → Jump
		8B06	10	
		8B07	8B	
	CPI 7AH	8B08	FE	Comp A with 'z' + 1
		8B09	7B	



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Label	Assembly Instruction	Memory Address (Hex)	Machine language (Hex)	Comment
	JNC false2	8B0A	D2	$c >= 'z' + 1 \rightarrow \text{Jump}$
		8B0B	10	
		8B0C	8B	$B \leftarrow 1$
	MVI B, 01H	8B0D	06	
		8B0E	01	
	RET	8B0F	C9	Return to caller
false2:	MVI B, 00H	8B10	06	$B \leftarrow 0$
		8B11	00	
	RET	8B12	C9	Return to caller
main:	MVI A, 62H	8900	3E	$A \leftarrow 'b' (\text{Test})$
	STA 8800H	8901	62	
	STA 8800H	8902	32	Store at 8800H
		8903	00	
		8904	88	
	CALL islower	8905	CD	Call subroutine
		8906	00	
		8907	8B	
	MOV A, B	8908	78	$A \leftarrow B (\text{return val})$
	STA 8802H	8909	32	Store at 8802H
		890A	02	
		890B	88	
	RST 3	890C	DF	Stop

Result: The test character 'b' (62H) is lowercase. After execution, checking memory location 8802H yields 01H.

Task 16: ~~Program~~ Program 11 (int isalpha(char c))

Action: Write a subroutine that uses CALL isupper and CALL islower. If either returns 1, isalpha returns 1.



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Label	Assembly Instruction	Memory Address (Hex)	Machine Language (Hex)	Comment
Parameters		8800H		Memory location for parameter c
isalpha:	CALL isupper	8C00	CD	Call 8A00H
		8C01	00	
		8C02	8A	
	MOV A,B	8C03	78	A ← B
	CPI 01H	8C04	FE	Is it upper? (1)
		8C05	01	
	SZ Done	8C06	CA	If 1 → Jump Done
		8C07	0C	
		8C08	8C	
	CALL islower	8C09	CD	Call 8B00H
		8C0A	00	
		8C0B	8B	
	RET	8C0C	C9	Return to caller
Done:				A ← '1' (Test)
main:	MVI A,31H	8900	3E	
		8901	31	
	STA 8800H	8902	32	Store at 8800H
		8903	00	
		8904	88	
	CALL isalpha	8905	CD	Call subroutine
		8906	00	
		8907	8C	
	MOV A,B	8908	78	A ← B (return val)
		8909	32	Store at 8802H
		890A	02	
		890B	88	
	RST 3	890C	DF	Stop



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-Result:-

1. Using the test character '1' (31H) which is non-alphabetical, memory location 8802H yields 00H.
2. Changing the test character to 'A' (41H) or 'z' (7AH) and re-running yields 01H at 8802H.

5. Challenges Faced

- Dealing with the 8085 limitation that comparisons can only be made against the Accumulator (Register A), which required constantly moving variables around.
- Calculating exact hex memory addresses for jump instructions to implement if-else and switch-case logic.
- Manually searching for the correct opcodes in the 2-page Instruction Set and reading through descriptions in the 16-page Instruction Set Reference Encyclopedia was a time-consuming process.

6. References

- 2-page Instruction Set Table for the 8085 Microprocessor.
- 16-page Instruction Set Reference Encyclopedia for the 8085 Microprocessor.
- Microprocessors Lab Report Guidelines July 2026.pdf (Laboratory Report Guidelines)

7. Report prepared by

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Group 16